

Evaluating Large Language Models for Arabic–English Ontology Matching

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Abstract

Large language models (LLMs) have demonstrated strong performance on ontology matching (OM) tasks, yet existing studies have focused almost exclusively on English-language ontologies. This paper presents a systematic evaluation of general-purpose LLMs for Arabic–English ontology matching using the MultiFarm benchmark, an established multilingual track of the Ontology Alignment Evaluation Initiative (OAEI). We evaluate three state-of-the-art models, GPT-4o, Claude 4.6, and Llama 3.3, and compare them with four established OAEI ontology matching systems under a zero-shot, full-candidate-list prompting strategy. Performance is evaluated using the standard OAEI precision, recall, and F_1 metrics. The results reveal a consistent pattern across all three LLMs: high performance on same-ontology Arabic–English matching (F_1 up to 0.89), but a substantial precision collapse on cross-ontology matching, where the models rarely predict NO_MATCH when no valid correspondence exists. These findings highlight the limitations of full-candidate-list prompting for heterogeneous ontology matching and motivate future work on retrieval-narrowed candidate generation for cross-lingual ontology matching.

Keywords

Ontology Matching, Large Language Models, Arabic, Cross-lingual Ontology Matching

1. Introduction

Ontology matching (OM) is a fundamental task in the Semantic Web that aims to identify correspondences between semantically equivalent entities across heterogeneous ontologies. Accurate ontology matching is essential for data integration, knowledge graph alignment, and interoperability across heterogeneous information systems. Recent studies have shown that large language models (LLMs) can effectively support ontology matching by leveraging contextual semantic knowledge to identify correspondences without task-specific training [1, 2].

Despite this progress, existing LLM-based ontology matching research has focused predominantly on English-language benchmarks. Cross-lingual ontology matching remains comparatively underexplored, particularly for Arabic–English ontology pairs. Arabic poses additional challenges due to its rich morphology and relatively limited representation in the multilingual training data of many LLMs, motivating a systematic evaluation of their effectiveness in Arabic–English ontology matching.

In this paper, we evaluate GPT-4o, Claude 4.6 Sonnet, and Llama 3.3 on the Arabic–English MultiFarm benchmark [3] using a zero-shot, full-candidate-list prompting strategy. We compare their performance with four established multilingual ontology matching systems. Our contributions are as follows:

- We provide the first systematic evaluation of three state-of-the-art LLMs on Arabic–English ontology matching with comparison to established multilingual OAEI matchers.
- We analyze the performance gap between same-ontology and cross-ontology matching, highlighting the remaining challenges of cross-lingual ontology matching across heterogeneous schemas.

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2. Related Work

Early work on applying LLMs to ontology matching framed the task mainly as pairwise classification, where the model determines whether two ontology entities are equivalent [4, 5]. Subsequent systems such as OLaLa [6], LLMs4OM [1], and Agent-OM [2] combined candidate retrieval with LLM-based verification to reduce the search space and improve matching precision. More recent retrieval-augmented approaches further enrich entity representations with contextual information before prompting [7]. However, most existing LLM-based OM studies focus primarily on English-language settings.

MultiFarm [3] is the established benchmark for multilingual ontology matching and contains translated Conference-domain ontologies in 12 languages, including Arabic. OAEI evaluations consistently show that cross-lingual matching is more challenging than monolingual matching [8]. Participating systems cover lexical, translation-based, structural, and hybrid approaches, including LogMap and LogMapLt [9, 10], LSMatch-Multilingual [11], and Matcha [12]. More recently, LogMapLLM [10] incorporated LLM reasoning into the LogMap pipeline, reflecting growing interest in hybrid LLM-based ontology matching. In contrast, our study evaluates zero-shot LLM prompting as a standalone matching strategy on the Arabic–English MultiFarm tasks.

Arabic introduces additional challenges for NLP because of its rich morphology, inflectional variation, and differences between Modern Standard Arabic and dialectal forms [13]. Recent benchmarks have evaluated LLMs on Arabic question answering, reasoning, and text generation tasks, showing substantial variation across models and task types [14, 13]. Arabic-focused models such as Jais [15] and AceGPT [16] have also been developed, but their effectiveness for ontology matching remains largely unexplored. To the best of our knowledge, this is the first systematic evaluation of general-purpose LLMs on Arabic–English ontology matching and the first analysis of their performance across same-ontology and cross-ontology settings.

3. Methodology

Figure 1 illustrates the proposed evaluation workflow. For each Arabic source ontology entity, the complete set of candidate English entity labels from the target ontology is provided to the LLM together with the prompt template. The model returns either the best matching candidate or NO_MATCH, and the resulting alignments are evaluated against the MultiFarm gold-standard reference alignments.

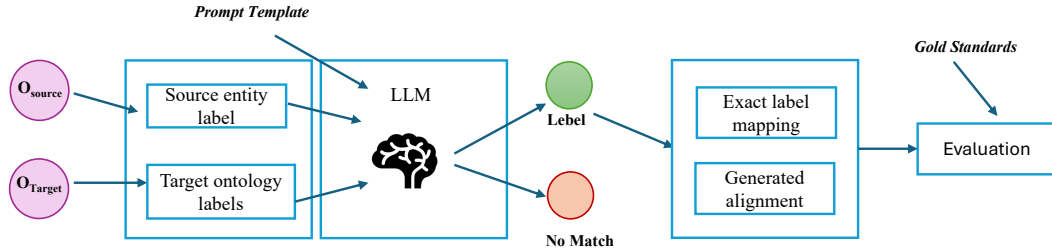


Figure 1: Overview of the LLM-based Arabic–English ontology matching workflow.

Datasets. We evaluate Arabic–English ontology matching using the MultiFarm benchmark [3], the multilingual extension of the OAEI Conference track. MultiFarm provides Arabic translations of five conference-domain ontologies—Cmt, Conference, ConfOf, Iasted, and Sigkdd—together with manually curated Arabic–English reference alignments. For the Arabic–English language pair, the benchmark defines 25 directional ontology matching tasks comprising 776 gold-standard correspondences.

We distinguish between two evaluation settings. **Same-ontology matching** aligns an ontology with its translated counterpart (e.g., Cmt-ar $\rightarrow$ Cmt-en). Since both ontologies share the same structure, this setting primarily evaluates the model’s ability to identify the correct English concept corresponding to

an Arabic label. **Cross-ontology matching** aligns different ontologies across languages (e.g., Cmt-ar → Conference-en). In this setting, the model must identify semantic equivalence across both a language boundary and a schema boundary, making the task substantially more challenging.

The benchmark therefore consists of five same-ontology tasks and twenty cross-ontology tasks, giving a total of twenty-five Arabic→English evaluation tasks.

Models. In order to evaluate the effectiveness of LLMs in Arabic-English ontology matching, we evaluate three state-of-the-art models: GPT-4o [17] (OpenAI), Claude 4.6 Sonnet [18] (Anthropic), and Llama 3.3 70B [19] (Meta, 70 billion parameters). These models were selected because of their strong multilingual capabilities, widespread adoption, and accessible inference APIs. Arabic-centric models such as Jais [15] and AceGPT [16] were considered but could not be included due to the lack of publicly accessible inference endpoints at the time of this study; evaluating such models remains an important direction for future work.

Prompting. Prompt design plays an important role in LLM-based ontology matching, as it influences the model’s understanding of the task and, consequently, its predictions. We adopt a zero-shot, full-candidate-list prompting strategy. For each Arabic source ontology entity, we construct a prompt containing (1) the Arabic label of the source entity and (2) the complete set of English labels from the target ontology as candidate matches. The model is instructed to select the single best-matching candidate or respond with NO_MATCH if no appropriate correspondence exists. To ensure consistent outputs, the model is instructed to return only a single candidate without any explanation. We deliberately use the full candidate list rather than a retrieval-narrowed shortlist to evaluate the model’s raw matching capability without retrieval-side assistance. Only ontology entity labels (i.e., classes and properties) are provided to the model. No ontology hierarchy, logical axioms, URIs, textual descriptions, or reference alignments are included in the prompt. This design intentionally evaluates the model’s ability to perform cross-lingual semantic matching using label information alone. The prompt template, used for all models and ontology pairs, is as follows:

You are an expert in ontology matching and bilingual (Arabic–English) terminology. Given an Arabic ontology concept and a list of candidate English ontology concepts from a different schema, identify which candidate (if any) refers to the same real-world concept as the Arabic term. Arabic concept: "{ar_label}"
Candidate English concepts:
{candidates_block}
Respond with only the exact text of the single best matching candidate, or respond with NO_MATCH if none of the candidates refer to the same concept. Do not explain your reasoning. Do not add punctuation or quotes around your answer.

The predicted English label was mapped to the corresponding ontology entity using exact string matching to generate the final alignment. Predictions of NO_MATCH produced no correspondence.

Baselines. We compare the LLMs with four multilingual ontology matching systems from the recent OAEI MultiFarm track: LogMap, LogMapLt, LSMatch-Multilingual, and Matcha [9, 11, 12]. These systems represent complementary matching paradigms, including lexical, translation-based, structural, and hybrid approaches, and provide strong baselines for assessing the effectiveness of zero-shot LLM prompting on Arabic–English ontology matching.

Experimental Settings. All experiments were conducted on an Apple M3 MacBook (16 GB RAM). GPT-4o and Claude were accessed through their respective APIs, while Llama 3.3 was accessed via the Groq inference API [20]. All models were evaluated in their standard configurations with temperature set to zero, without fine-tuning or task-specific training.

For comparison, LogMap, LogMapLt, LSMatch-Multilingual, and Matcha were executed on the same Arabic–English MultiFarm tasks using the MELT evaluation client [21].

The complete implementation, prompt templates, generated alignments, and evaluation scripts are publicly available in the accompanying GitHub repository.¹

Evaluation. The generated alignments are evaluated against the MultiFarm gold-standard reference alignments using Precision, Recall, and F_1 -score. Precision measures the proportion of predicted correspondences that are correct, Recall measures the proportion of reference correspondences that are successfully identified, and F_1 is the harmonic mean of Precision and Recall. For each evaluation setting (same-ontology and cross-ontology), the reported Precision, Recall, and F_1 values are obtained by macro-averaging the corresponding metric across all tasks within that setting.

4. Results

Table 1 reports the macro-averaged Precision (P), Recall (R), and F_1 scores of the evaluated LLMs and OAEI matchers on the 25 Arabic–English MultiFarm ontology matching tasks. Results are reported separately for the five same-ontology tasks, the twenty cross-ontology tasks, and overall. In our evaluation, LogMapLt and Matcha produced no correspondences on any of the Arabic–English MultiFarm tasks. Consequently, their precision, recall, and F_1 scores were all zero.

Table 1

Macro-averaged results for LLMs and OAEI 2025 matchers on the MultiFarm ar–en track (25 pairs). Same-onto: 5 same-ontology pairs. Cross-onto: 20 cross-ontology pairs.

System	Same-onto (5)			Cross-onto (20)			Overall (25)		
	P	R	F_1	P	R	F_1	P	R	F_1
Large Language Models									
Claude 4.6 [18]	0.884	0.900	0.892	0.113	0.805	0.198	0.267	0.824	0.403
GPT-4o [17]	0.833	0.839	0.836	0.107	0.741	0.187	0.252	0.760	0.379
Llama 3.3 [19]	0.736	0.726	0.731	0.106	0.668	0.183	0.232	0.680	0.346
OAEI2025 matchers									
LSMatch-ML [11]	0.943	0.511	0.663	0.769	0.535	0.631	0.803	0.530	0.639
LogMap [9]	0.955	0.433	0.595	0.748	0.318	0.446	0.789	0.341	0.476
LogMapLt [9]	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Matcha [12]	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

On same-ontology tasks, the LLMs achieve substantially higher F_1 scores than the traditional OAEI matchers. Claude achieves the highest F_1 of 0.892, followed by GPT-4o ($F_1=0.836$) and Llama 3.3 ($F_1=0.731$). By contrast, the best OAEI matcher, LSMatch-ML, achieves $F_1=0.663$, and LogMap achieves $F_1=0.595$. The performance gap is driven primarily by recall: LLMs recover 73–90% of true correspondences compared to 43–51% for the OAEI matchers, indicating that LLMs are substantially better at recognising semantically equivalent Arabic–English concept pairs when the schema structure is shared. Precision is comparable between LLMs (0.74–0.88) and OAEI matchers (0.94–0.96), though traditional matchers remain slightly more conservative. In the same-ontology setting, corresponding entities share the same ontology structure and typically differ only in language, making the task closely resemble cross-lingual lexical matching. Cross-ontology matching, by contrast, requires recognising semantic equivalence across both a language boundary and heterogeneous conceptualisations.

Under this more challenging setting, the traditional matchers outperform all three LLMs in terms of F_1 : LSMatch-ML achieves $F_1=0.631$ and LogMap $F_1=0.446$, while Claude, GPT-4o, and Llama 3.3 achieve $F_1=0.198$, 0.187, and 0.183 respectively. The failure mode is consistent across all three models:

¹<https://github.com/OmaimaFallatah/llm-arabic-english-om>

recall remains high (0.67–0.81), but precision collapses to 0.11, meaning the models produce large numbers of false correspondences. In the full-candidate-list prompting setup, the LLM is presented with the entire target ontology (75–182 concepts) and must identify which, if any, corresponds to the Arabic source concept. Although the prompt instructs the models to return `NO_MATCH`, when no true correspondence exists, the models rarely predict `NO_MATCH`. They instead select the most semantically plausible candidate from the provided list, producing false positives at high rates. This behaviour is illustrated by the `iasted-ar` to `cmt-en` task, where Claude generated 94 correspondences despite the gold standard containing only four. This “forced choice” behaviour under full-candidate prompting is the primary driver of the precision collapse observed across all cross-ontology pairs. LSMatch-ML’s stronger cross-ontology precision (0.769) can be attributed to its translation-memory-based approach: it only adds a correspondence when a translated Arabic label closely matches an English string, producing fewer but more reliable alignments. LLMs, by contrast, reason semantically across the full candidate list and tend to propose plausible but incorrect matches when no true correspondence exists.

In terms of LLMs’ ability to match Arabic-English ontologies, Claude consistently achieves the highest scores across both matching conditions, followed by GPT-4o and Llama 3.3. This suggests that general multilingual capability, rather than Arabic-specific training, is the primary driver of performance on this task. Moreover, the performance gap between models is more pronounced on same-ontology matching, F_1 ranges from 0.731 to 0.892, than on cross-ontology matching, F_1 ranges from 0.183–0.198, where all three models yield similarly low precision regardless of their overall capability. These results indicate that strong general multilingual capability is sufficient to achieve competitive performance on this task, although evaluating Arabic-specific LLMs remains an important direction for future work.

5. Conclusion and Future Work

This paper presented a systematic evaluation of general-purpose LLMs for Arabic-English ontology matching using the MultiFarm benchmark. We evaluated three state-of-the-art models, GPT-4o, Claude, and Llama 3.3, under a zero-shot and full-candidate-list prompting strategy. We compared LLMs’ results against four established OAEI matchers: LSMatch-Multilingual, LogMap, LogMapLt, and Matcha. Our results reveal a clear distinction between same-ontology and cross-ontology matching. LLMs substantially outperform OAEI matchers, with Claude achieving $F_1=0.892$ compared to LSMatch-ML’s $F_1=0.663$, driven by significantly higher recall. On cross-ontology matching, the situation reverses: traditional matchers, particularly LSMatch-ML, outperform all three LLMs. The precision of all three LLMs collapses because the models exhibit a “forced choice” behaviour under full-candidate-list prompting, selecting semantically plausible candidates even when no correct correspondence exists. These findings suggest two directions for future work. First, retrieval-narrowed prompting strategies, where the candidate list is filtered to a small set of semantically plausible matches before querying the LLM. This could substantially improve on cross-ontology precision while preserving the high recall. Second, evaluating Arabic-centric models which could not be included in this study due to infrastructure constraints, remains an open and important question. Larger multilingual datasets should also be explored to develop a more comprehensive framework for Arabic-aware cross-lingual ontology matching.

Declaration on Generative AI

During the preparation of this work, the author used ChatGPT-5.5 and Claude in order to: Grammar and spelling check. After using these tools/services, the author reviewed and edited the content as needed and takes full responsibility for the publication’s content.

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